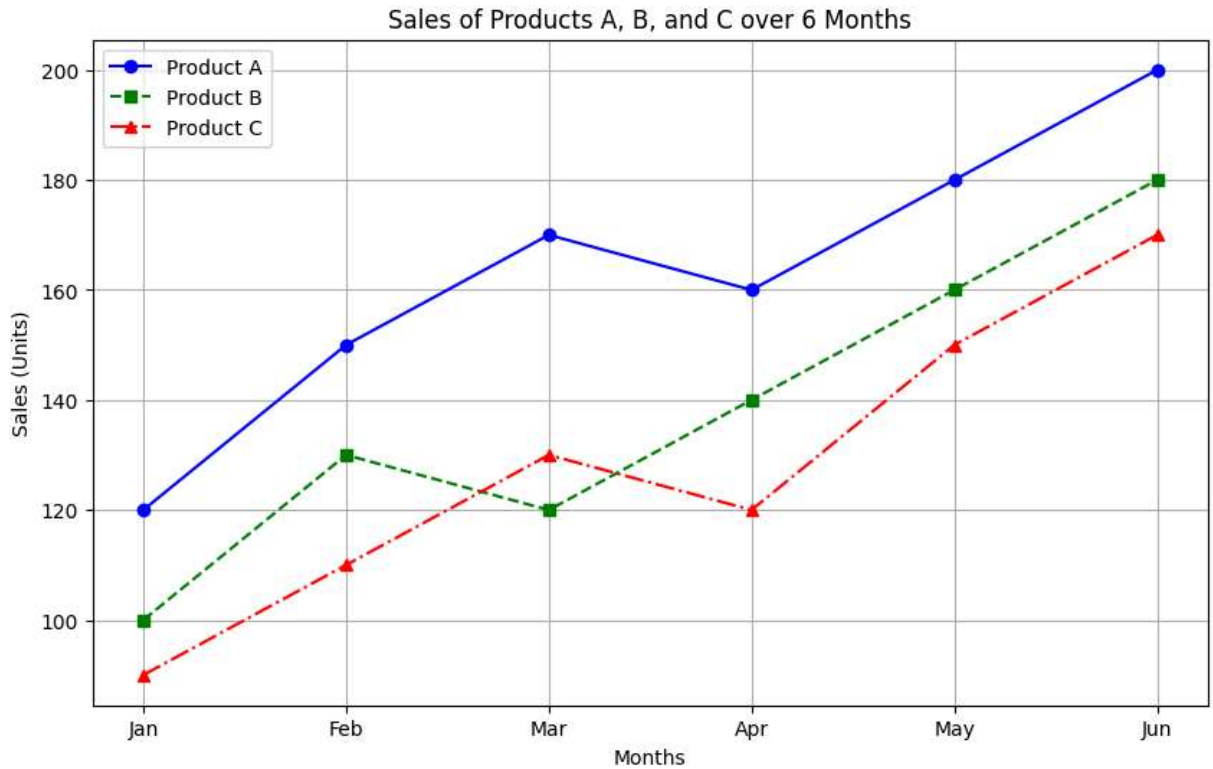
In [1]:
```
import matplotlib.pyplot as plt

# Months
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
          "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]

# Monthly sales in units (example data)
sales = [250, 300, 280, 350, 400, 420, 380, 390, 450, 470, 430, 500]

# Create the line chart
plt.figure(figsize=(10,6))
plt.plot(months, sales, marker='o', linestyle='-', color='blue', label="Sales")
plt.title("Monthly Sales of the Company")
plt.xlabel("Months")
plt.ylabel("Sales (Units)")
plt.grid(True)
plt.legend()
plt.show()
```

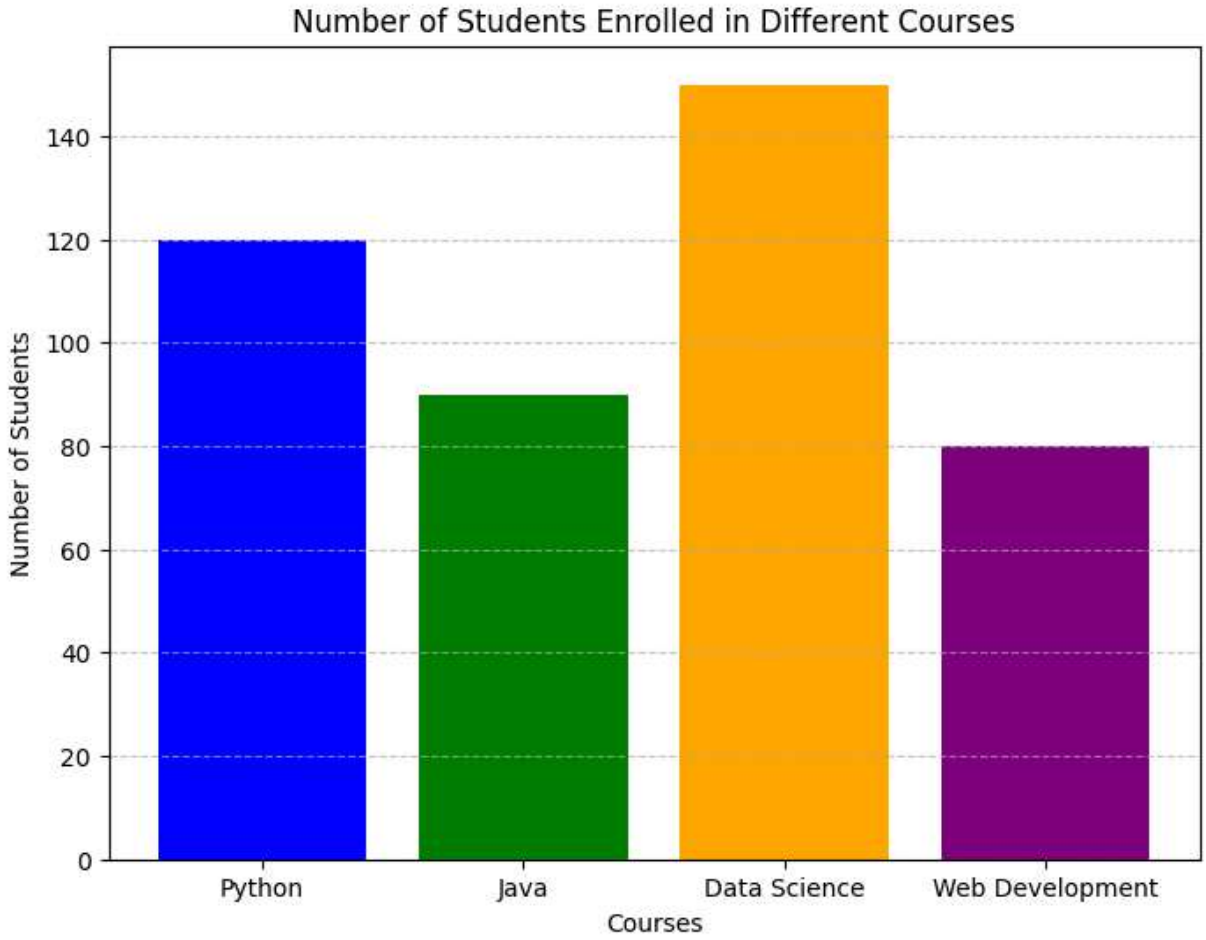
In [2]:
```
import matplotlib.pyplot as plt

# Months
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]

# Sales data for three products
sales_A = [120, 150, 170, 160, 180, 200]
sales_B = [100, 130, 120, 140, 160, 180]
sales_C = [90, 110, 130, 120, 150, 170]

# Plotting the lines
```

```
plt.figure(figsize=(10,6))
plt.plot(months, sales_A, marker='o', linestyle='-', color='blue', label="Product A
plt.plot(months, sales_B, marker='s', linestyle='--', color='green', label="Product
plt.plot(months, sales_C, marker='^', linestyle='-.', color='red', label="Product C

# Adding title and labels
plt.title("Sales of Products A, B, and C over 6 Months")
plt.xlabel("Months")
plt.ylabel("Sales (Units)")
plt.grid(True)
plt.legend()
plt.show()
```

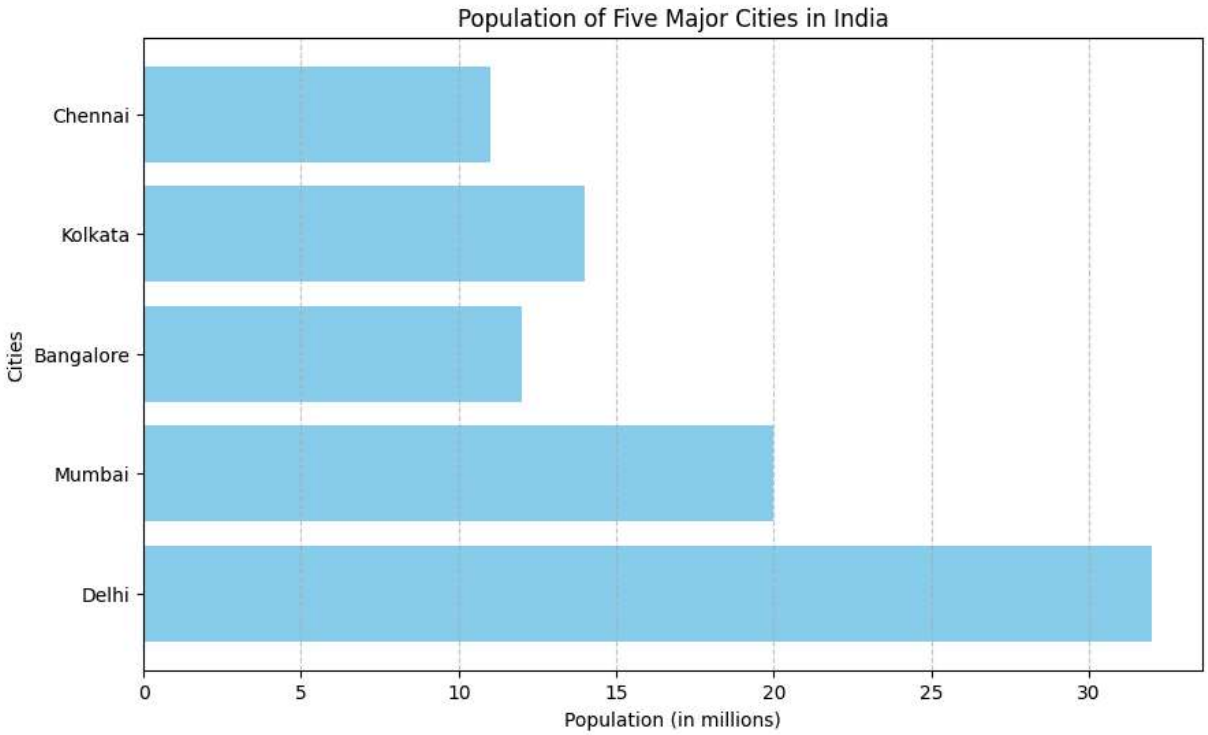



In [3]:

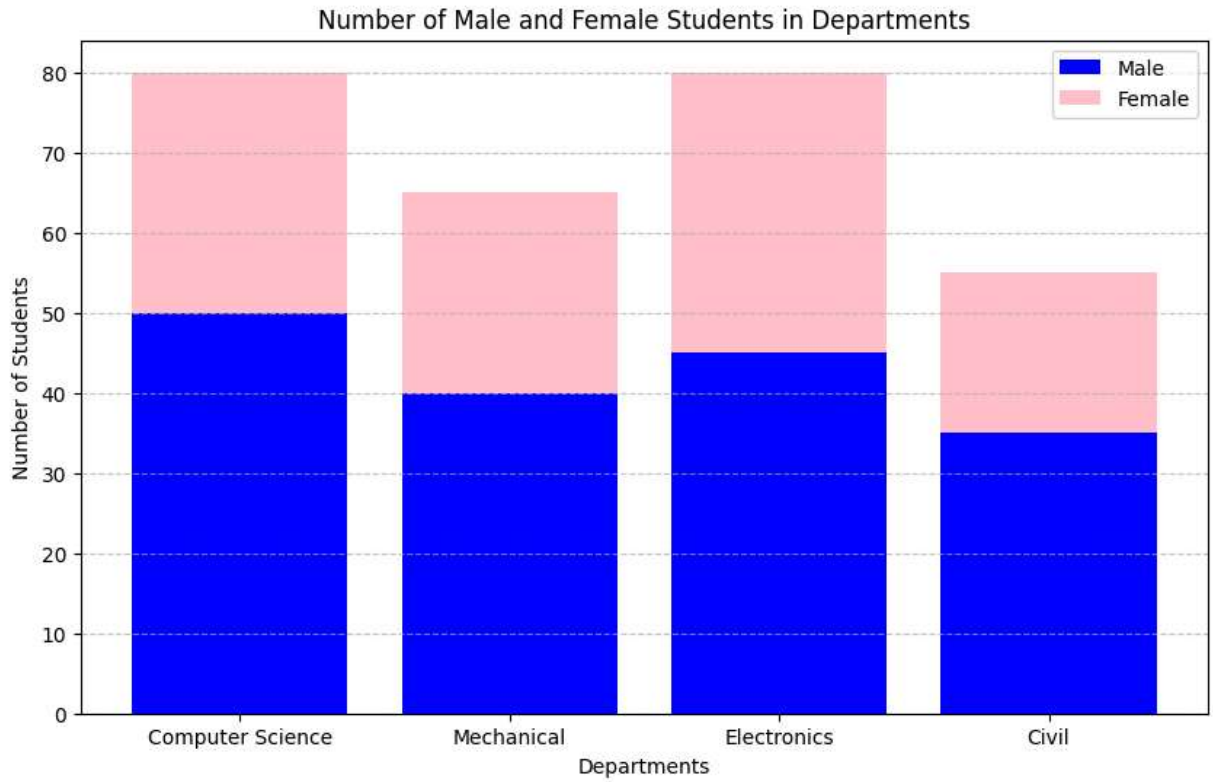
```
import matplotlib.pyplot as plt

# Courses
courses = ["Python", "Java", "Data Science", "Web Development"]

# Number of students enrolled (example data)
students = [120, 90, 150, 80]

# Create the bar chart
plt.figure(figsize=(8,6))
plt.bar(courses, students, color=['blue', 'green', 'orange', 'purple'])

# Adding title and labels
plt.title("Number of Students Enrolled in Different Courses")
plt.xlabel("Courses")
plt.ylabel("Number of Students")
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```



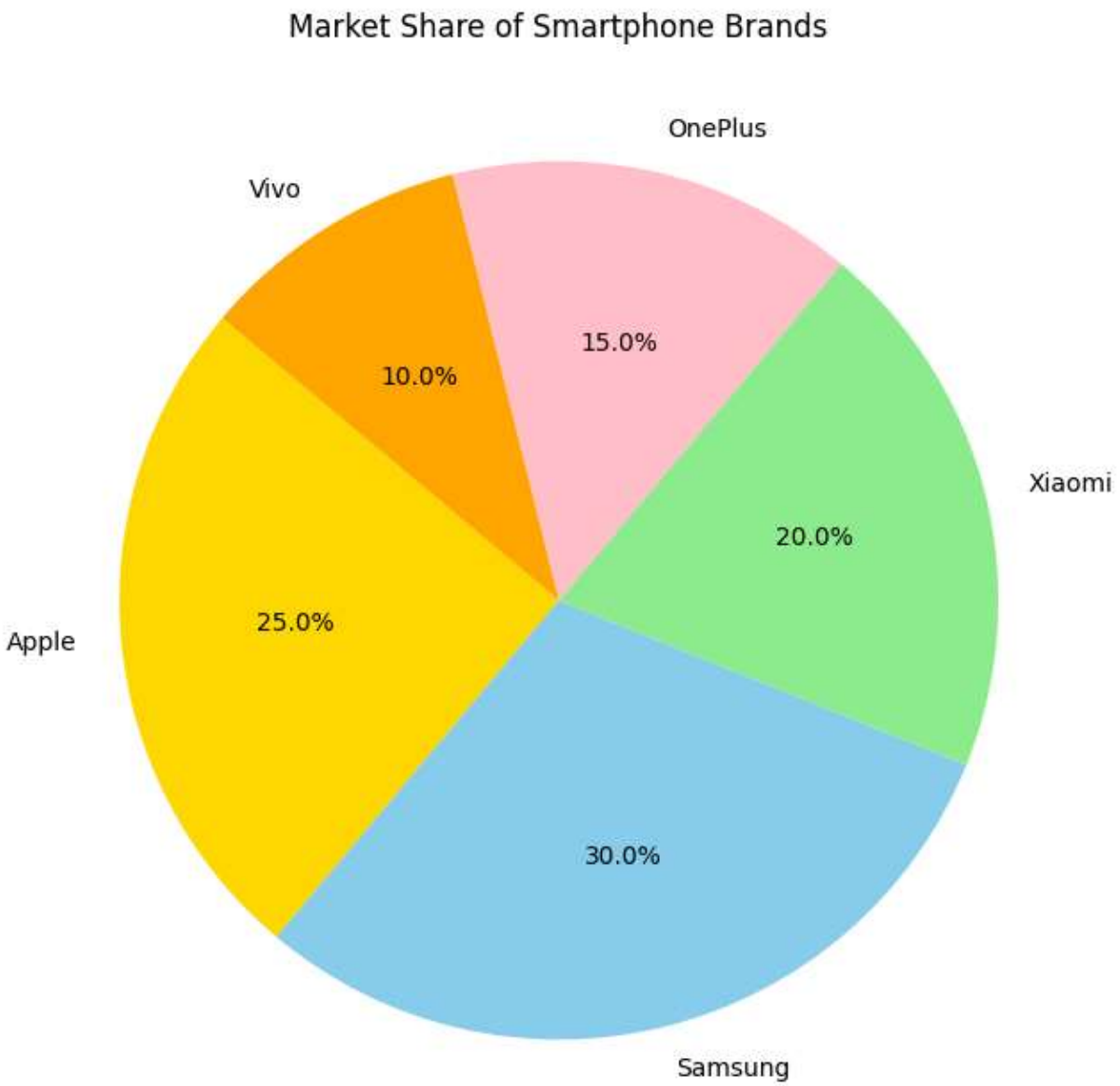

In [4]:
```
import matplotlib.pyplot as plt

# Cities
cities = ["Delhi", "Mumbai", "Bangalore", "Kolkata", "Chennai"]

# Population in millions (example data)
population = [32, 20, 12, 14, 11]

# Create horizontal bar chart
plt.figure(figsize=(10,6))
plt.barh(cities, population, color='skyblue')
plt.xlabel("Population (in millions)")
plt.ylabel("Cities")
plt.title("Population of Five Major Cities in India")
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.show()
```



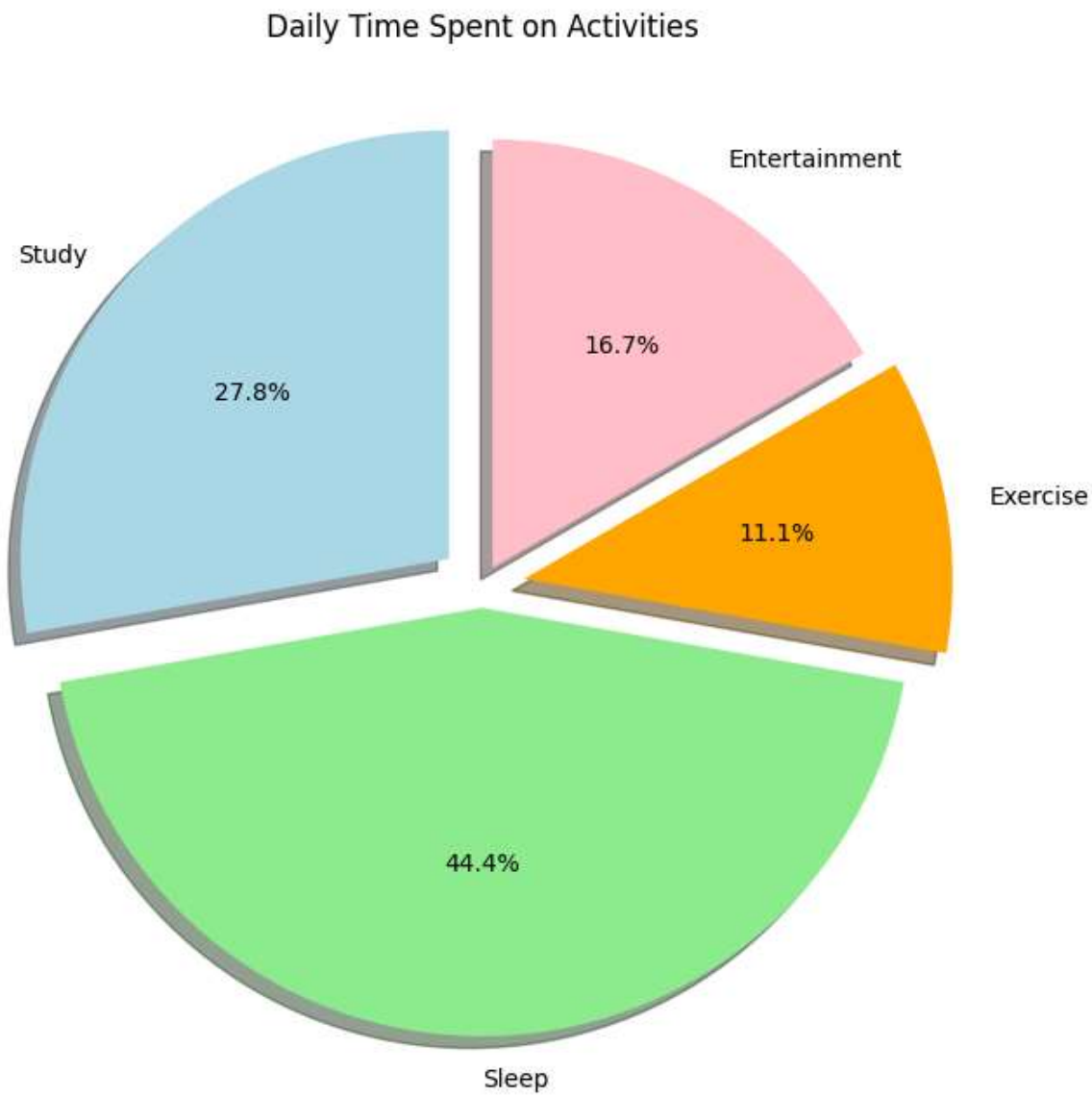

In [5]:
```
import matplotlib.pyplot as plt

# Departments
departments = ["Computer Science", "Mechanical", "Electronics", "Civil"]

# Number of male and female students (example data)
male_students = [50, 40, 45, 35]
female_students = [30, 25, 35, 20]

# Create stacked bar chart
plt.figure(figsize=(10,6))
plt.bar(departments, male_students, color='blue', label='Male')
plt.bar(departments, female_students, bottom=male_students, color='pink', label='Fe

# Adding title and labels
plt.title("Number of Male and Female Students in Departments")
plt.xlabel("Departments")
plt.ylabel("Number of Students")
plt.legend()
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



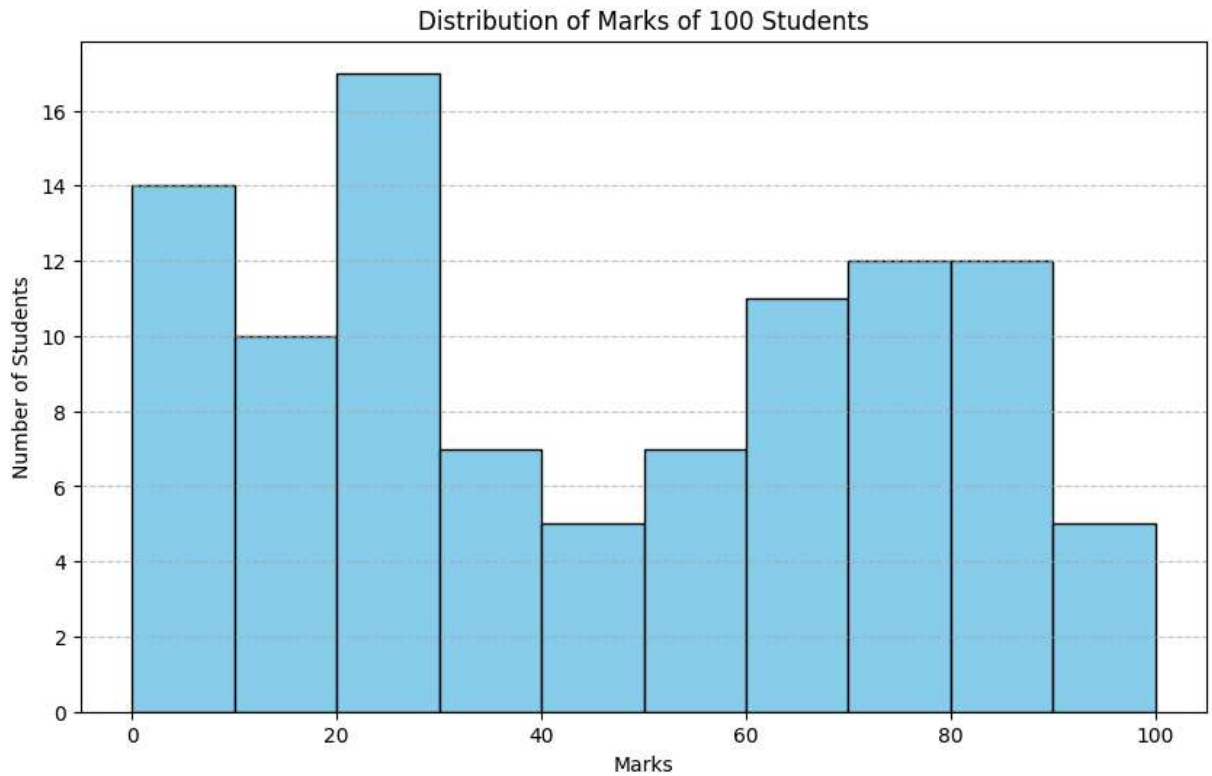

In [6]:
```
import matplotlib.pyplot as plt

# Smartphone brands
brands = ["Apple", "Samsung", "Xiaomi", "OnePlus", "Vivo"]

# Market share in percentage (example data)
market_share = [25, 30, 20, 15, 10]

# Create pie chart
plt.figure(figsize=(8,8))
plt.pie(market_share, labels=brands, autopct='%1.1f%%', startangle=140, colors=['go
plt.title("Market Share of Smartphone Brands")
plt.show()
```



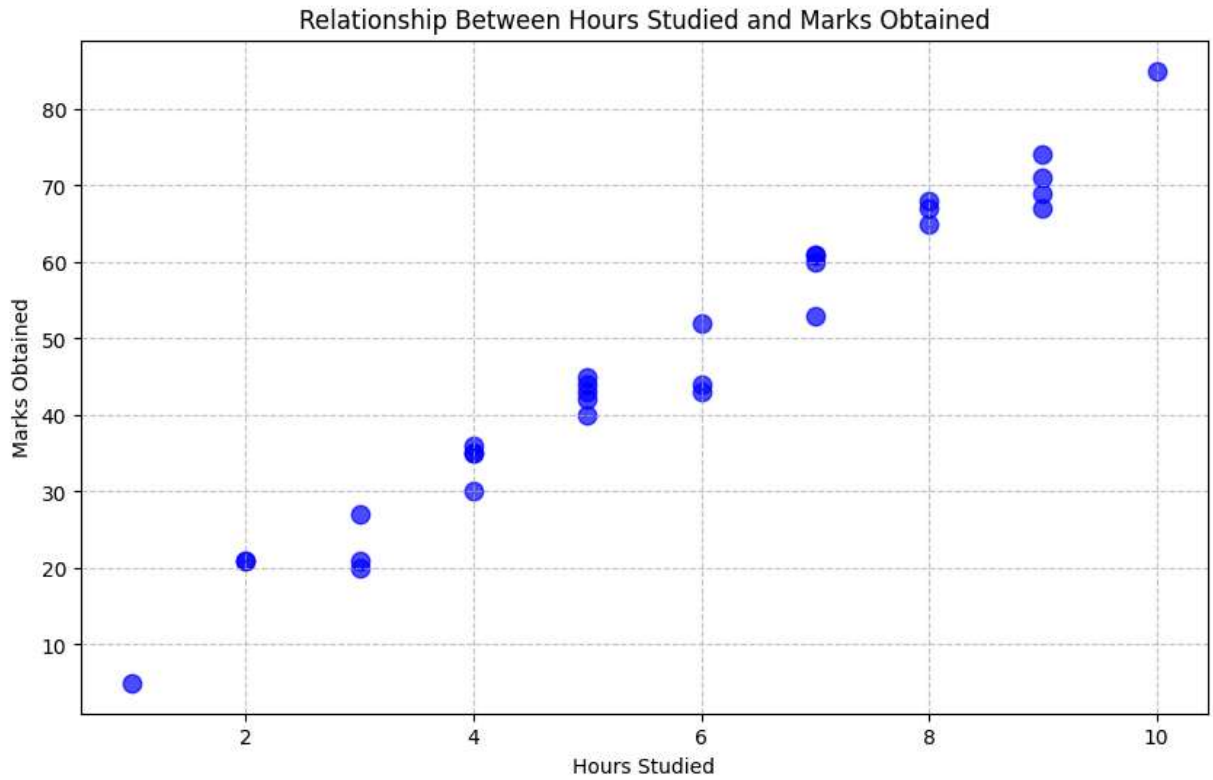

In [7]:
```
import matplotlib.pyplot as plt

# Activities
activities = ["Study", "Sleep", "Exercise", "Entertainment"]

# Time spent in hours (example data)
time_spent = [5, 8, 2, 3]

# Explode values to separate each slice slightly
explode = [0.1, 0.05, 0.1, 0.05]  # adjust for emphasis

# Create exploded pie chart
plt.figure(figsize=(8,8))
plt.pie(time_spent, labels=activities, autopct='%1.1f%%', startangle=90, explode=ex
        shadow=True, colors=['lightblue', 'lightgreen', 'orange', 'pink'])
plt.title("Daily Time Spent on Activities")
plt.show()
```



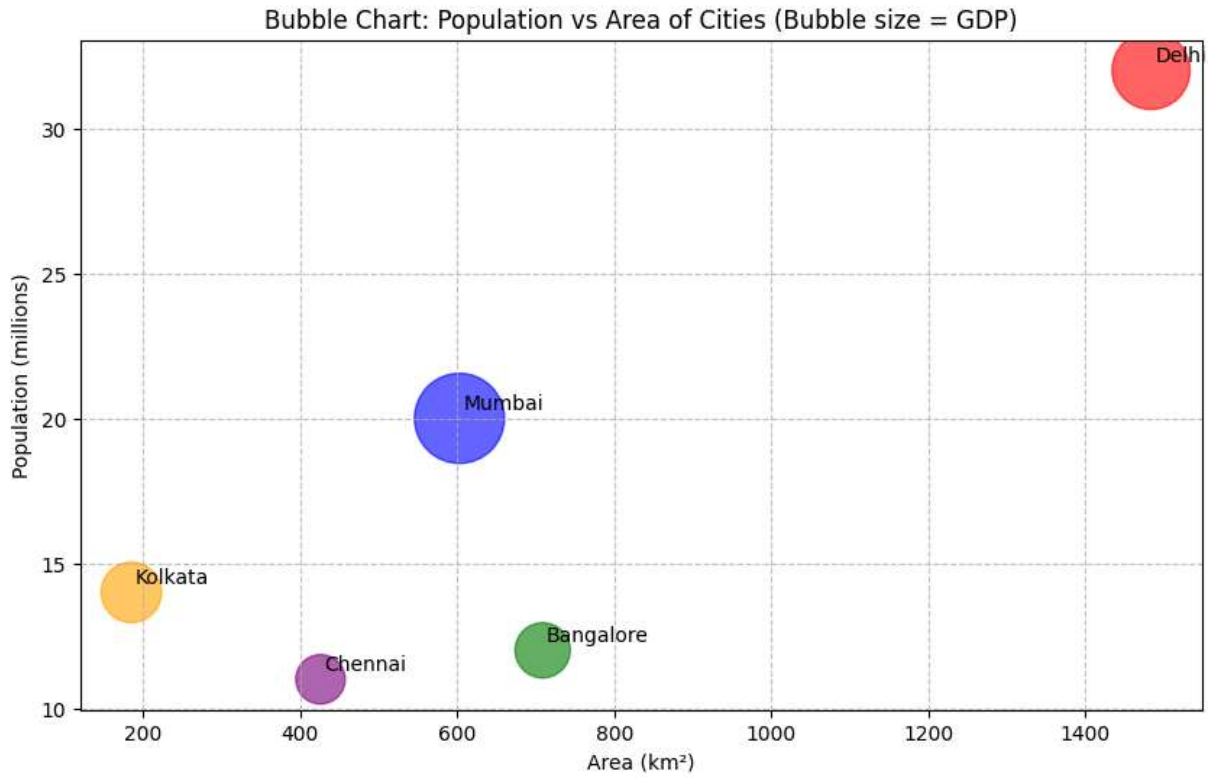

In [8]:
```
import matplotlib.pyplot as plt
import numpy as np

# Generate random marks for 100 students (0 to 100)
np.random.seed(1)  # for reproducibility
marks = np.random.randint(0, 101, 100)

# Create histogram
plt.figure(figsize=(10,6))
plt.hist(marks, bins=10, color='skyblue', edgecolor='black')  # 10 bins for ranges
plt.title("Distribution of Marks of 100 Students")
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



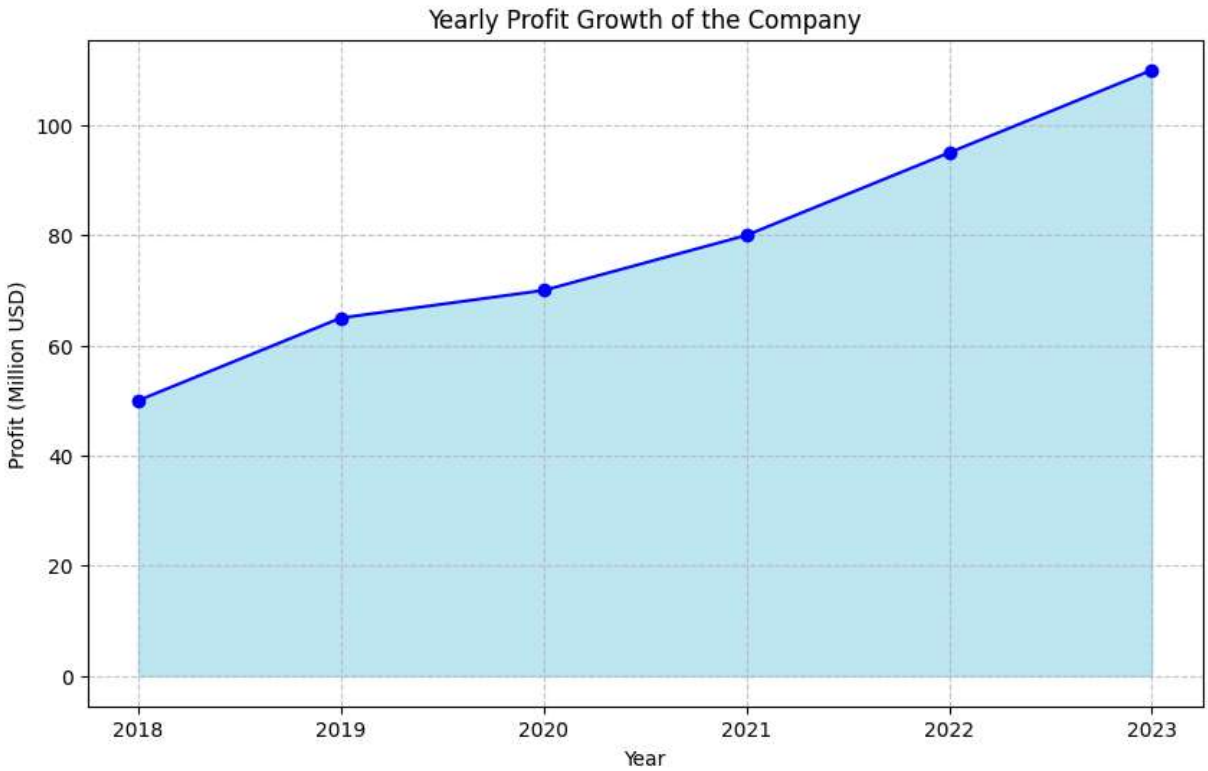

In [9]:
```
import matplotlib.pyplot as plt
import numpy as np

# Example data: hours studied and marks obtained
np.random.seed(2)  # for reproducibility
hours_studied = np.random.randint(1, 11, 30)  # 30 students studied 1 to 10 hours
marks_obtained = hours_studied * 8 + np.random.randint(-5, 6, 30)  # marks roughly

# Create scatter plot
plt.figure(figsize=(10,6))
plt.scatter(hours_studied, marks_obtained, color='blue', s=80, alpha=0.7)  # s=mark
plt.title("Relationship Between Hours Studied and Marks Obtained")
plt.xlabel("Hours Studied")
plt.ylabel("Marks Obtained")
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



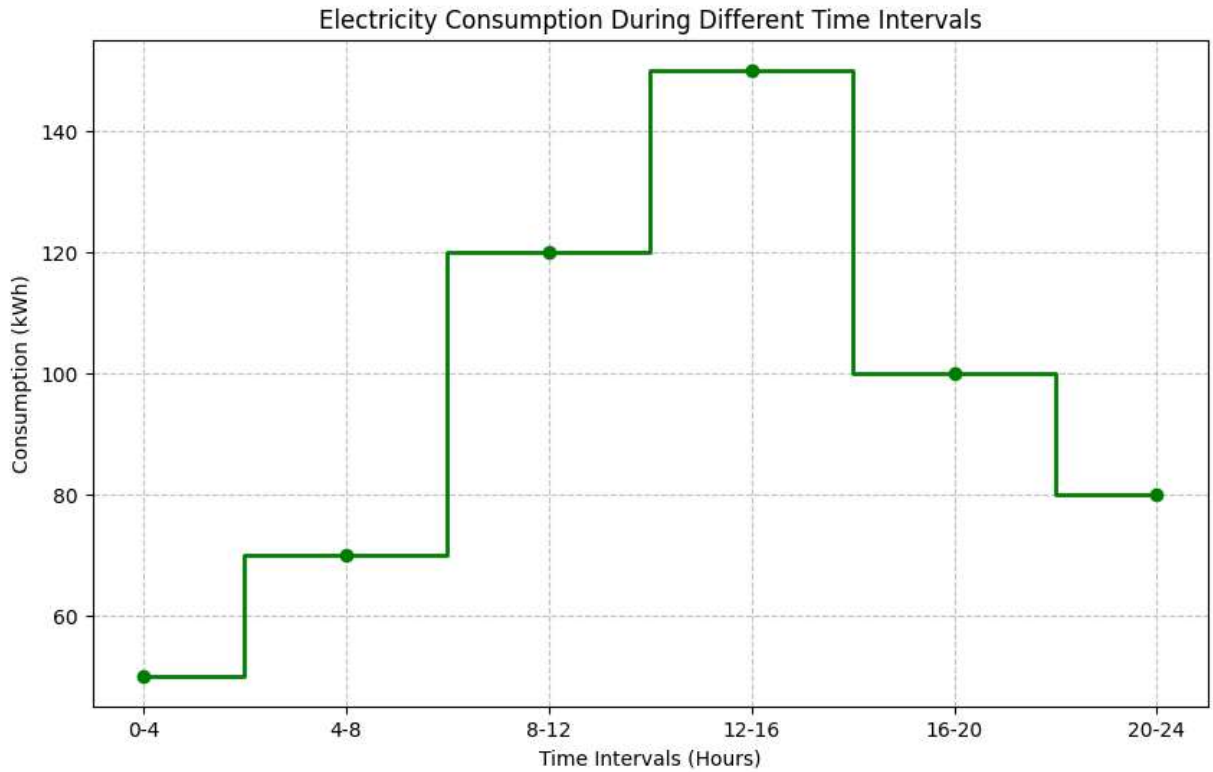

In [10]:
```
import matplotlib.pyplot as plt

# Example data for 5 cities
cities = ["Delhi", "Mumbai", "Bangalore", "Kolkata", "Chennai"]
area = [1484, 603, 709, 185, 426]       # Area in km²
population = [32, 20, 12, 14, 11]       # Population in millions
gdp = [300, 400, 150, 180, 120]         # GDP in billion USD

# Bubble sizes (scaled for visualization)
bubble_size = [g*5 for g in gdp]  # multiply to make bubbles visible

# Create bubble chart
plt.figure(figsize=(10,6))
plt.scatter(area, population, s=bubble_size, alpha=0.6, c=['red','blue','green','or
plt.title("Bubble Chart: Population vs Area of Cities (Bubble size = GDP)")
plt.xlabel("Area (km²)")
plt.ylabel("Population (millions)")

# Add labels for each city
for i, city in enumerate(cities):
    plt.text(area[i]+5, population[i]+0.3, city)

plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



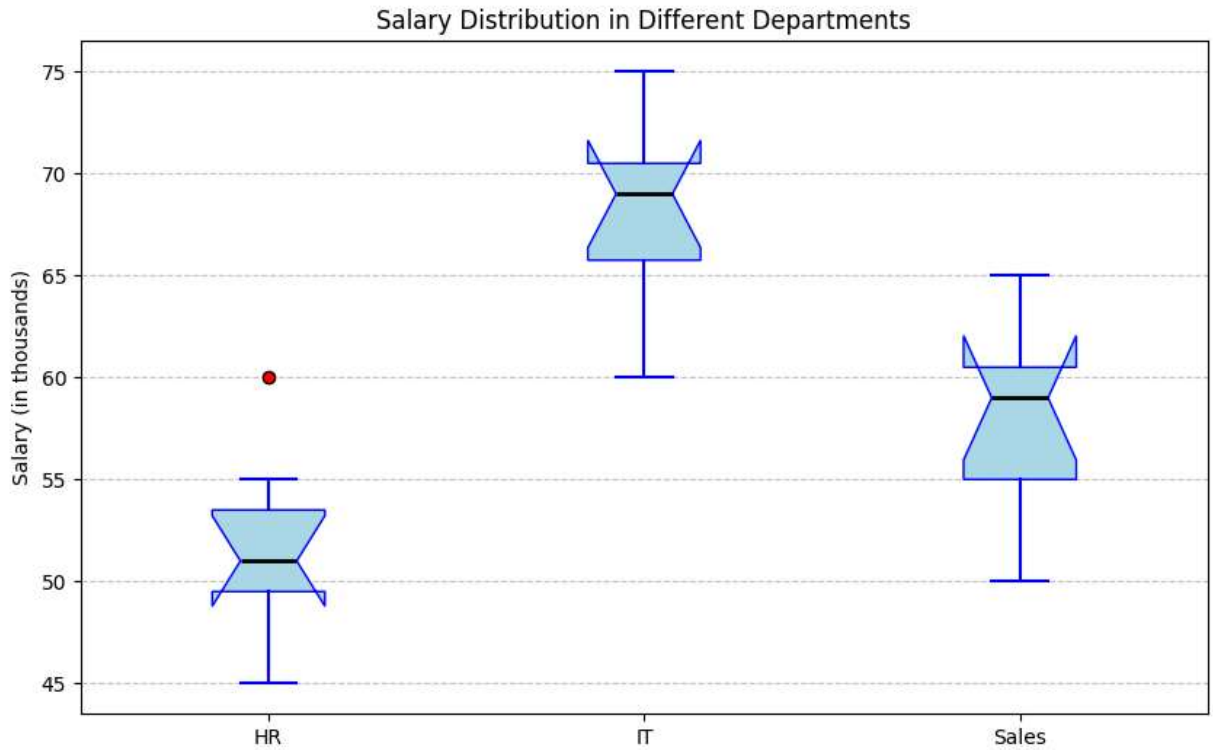

In [11]:
```
import matplotlib.pyplot as plt

# Years
years = [2018, 2019, 2020, 2021, 2022, 2023]

# Profit in million USD (example data)
profit = [50, 65, 70, 80, 95, 110]

# Create area chart
plt.figure(figsize=(10,6))
plt.fill_between(years, profit, color='skyblue', alpha=0.5)  # fill area under the
plt.plot(years, profit, color='blue', marker='o', linestyle='-')  # overlay line fo
plt.title("Yearly Profit Growth of the Company")
plt.xlabel("Year")
plt.ylabel("Profit (Million USD)")
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



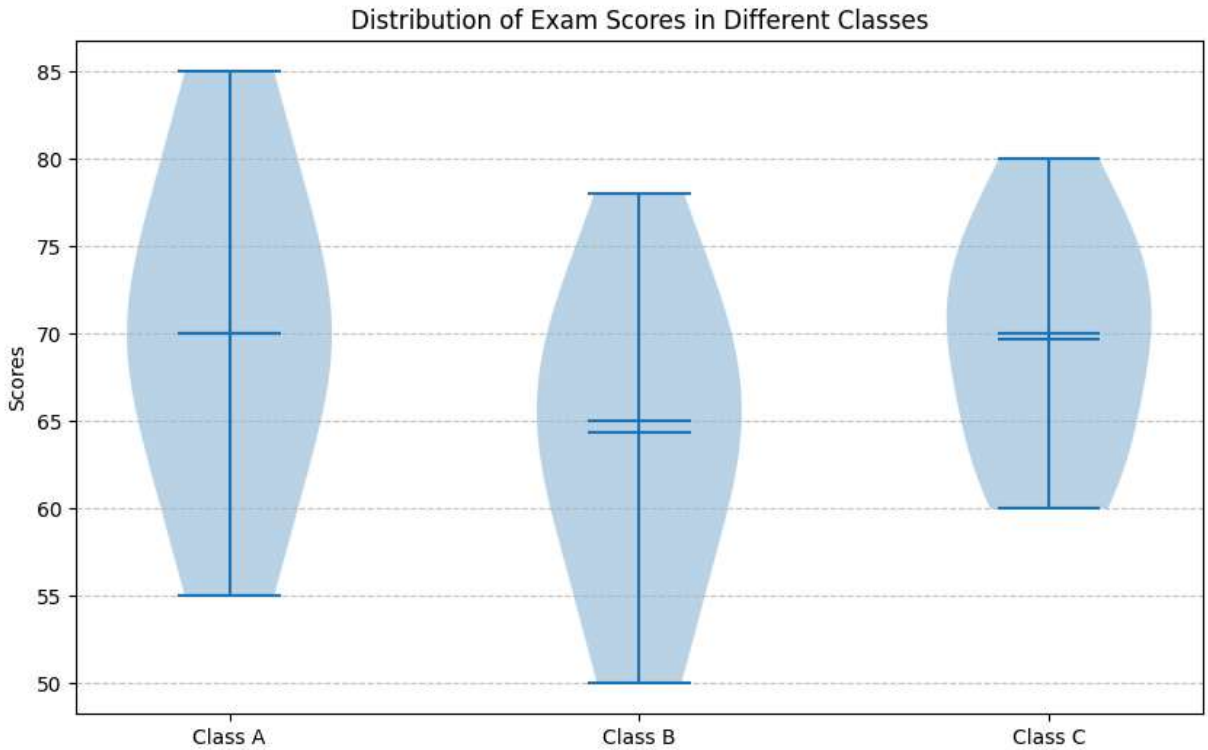

In [14]:
```
import matplotlib.pyplot as plt

# Time intervals (in hours)
time_intervals = ["0-4", "4-8", "8-12", "12-16", "16-20", "20-24"]

# Electricity consumption in kWh (example data)
consumption = [50, 70, 120, 150, 100, 80]

# Create step chart
plt.figure(figsize=(10,6))
plt.step(time_intervals, consumption, where='mid', marker='o', color='green', linew
plt.title("Electricity Consumption During Different Time Intervals")
plt.xlabel("Time Intervals (Hours)")
plt.ylabel("Consumption (kWh)")
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```




In [16]:
```
import matplotlib.pyplot as plt

# Salary data in thousands
HR_salaries = [45, 50, 55, 60, 50, 52, 48, 53]
IT_salaries = [60, 65, 70, 75, 68, 72, 66, 70]
Sales_salaries = [50, 55, 60, 65, 58, 62, 55, 60]

# Combine data
salary_data = [HR_salaries, IT_salaries, Sales_salaries]

# Department names
departments = ["HR", "IT", "Sales"]

# Create box plot
plt.figure(figsize=(10,6))
plt.boxplot(
    salary_data,
    patch_artist=True,        # Fill boxes with color
    notch=True,               # Show notch for median
    tick_labels=departments,  # Updated parameter to avoid warning
    boxprops=dict(facecolor='lightblue', color='blue'),  # Box color
    medianprops=dict(color='black', linewidth=2),       # Median line in black
    whiskerprops=dict(color='blue', linewidth=1.5),
    capprops=dict(color='blue', linewidth=1.5),
    flierprops=dict(marker='o', markerfacecolor='red', markersize=6, linestyle='non
)

plt.title("Salary Distribution in Different Departments")
plt.ylabel("Salary (in thousands)")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

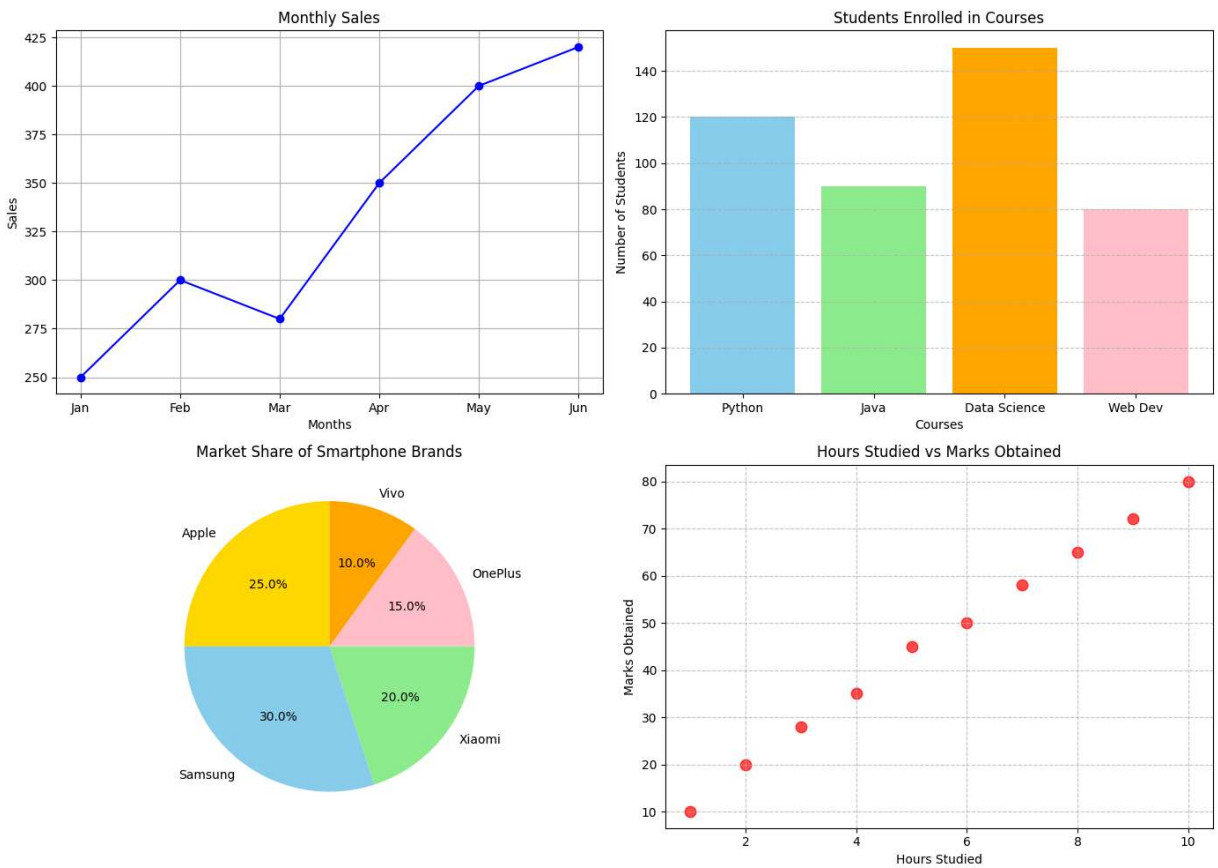



In [17]:
```
import matplotlib.pyplot as plt

# Example exam scores for 3 classes
class_A = [55, 60, 65, 70, 72, 68, 75, 80, 85]
class_B = [50, 55, 60, 65, 70, 62, 67, 72, 78]
class_C = [60, 62, 65, 68, 70, 72, 74, 76, 80]

# Combine data
scores = [class_A, class_B, class_C]

# Class labels
classes = ["Class A", "Class B", "Class C"]

# Create violin plot
plt.figure(figsize=(10,6))
plt.violinplot(scores, showmeans=True, showmedians=True)
plt.xticks([1, 2, 3], classes)
plt.title("Distribution of Exam Scores in Different Classes")
plt.ylabel("Scores")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```




In [18]:
```
import matplotlib.pyplot as plt

# Sample data

# Line Chart: Monthly sales
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [250, 300, 280, 350, 400, 420]

# Bar Chart: Students in courses
courses = ["Python", "Java", "Data Science", "Web Dev"]
students = [120, 90, 150, 80]

# Pie Chart: Smartphone market share
brands = ["Apple", "Samsung", "Xiaomi", "OnePlus", "Vivo"]
market_share = [25, 30, 20, 15, 10]

# Scatter Plot: Hours studied vs Marks obtained
hours_studied = [1,2,3,4,5,6,7,8,9,10]
marks_obtained = [10, 20, 28, 35, 45, 50, 58, 65, 72, 80]

# Create a figure with 2x2 subplots
fig, axs = plt.subplots(2, 2, figsize=(14,10))

# --- Line Chart ---
axs[0,0].plot(months, sales, marker='o', color='blue', linestyle='-')
axs[0,0].set_title("Monthly Sales")
axs[0,0].set_xlabel("Months")
axs[0,0].set_ylabel("Sales")
axs[0,0].grid(True)

# --- Bar Chart ---
axs[0,1].bar(courses, students, color=['skyblue', 'lightgreen', 'orange', 'pink'])
```

```
axs[0,1].set_title("Students Enrolled in Courses")
axs[0,1].set_xlabel("Courses")
axs[0,1].set_ylabel("Number of Students")
axs[0,1].grid(axis='y', linestyle='--', alpha=0.7)

# --- Pie Chart ---
axs[1,0].pie(market_share, labels=brands, autopct='%1.1f%%', startangle=90,
             colors=['gold', 'skyblue', 'lightgreen', 'pink', 'orange'])
axs[1,0].set_title("Market Share of Smartphone Brands")

# --- Scatter Plot ---
axs[1,1].scatter(hours_studied, marks_obtained, color='red', s=80, alpha=0.7)
axs[1,1].set_title("Hours Studied vs Marks Obtained")
axs[1,1].set_xlabel("Hours Studied")
axs[1,1].set_ylabel("Marks Obtained")
axs[1,1].grid(True, linestyle='--', alpha=0.7)

# Adjust layout to prevent overlap
plt.tight_layout()
plt.show()
```




In [ ]: